Task 7: Data Understanding and Measurement Basics

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Objective

The objective of this task is to build the theoretical base required for data analysis before moving into pandas, feature creation, visualization, or machine learning. Participants must understand what a dataset represents, how measurements are structured, how variables differ, and why domain context and units matter before analysis.

Understanding Data: Observations vs Variables

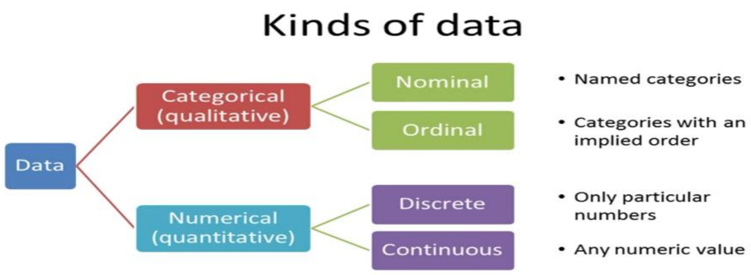
Data is typically structured as a table or a “Dataframe” which is made up of rows and columns. Each observation corresponds to one single instance in the dataframe, such as one sensor reading or one experimental trial. Each variable is an attribute which is measured across all observations, ie. A specific property like temperature, voltage, pH etc. In tabular form, observations correspond to the rows and the variables correspond to the columns.

Types and Nature of Data

- **Categorical vs Numerical Data**

  Data can be split into two broad types, categorical and numerical. Categorical data, as the name implies defines categories whereas numerical data is strictly quantitative. Categorical data can be further divided into nominal(named categories like colors of cars : red, green blue etc.) and ordinal data(categories with implied order like big, medium and small). Numerical data can also be broken into two kinds Discrete and Continuous, Discrete data will have only a fixed array of numeric values whereas continuous data will contain any numeric value.

Domain	Categorical Data	Numeric Data
Biochemistry	Biomolecule Types (Proteins, Nucleic Acids, Carbohydrates, Lipids)	Molecular Composition Counts: Number of base pairs in a plasmid, number of amino acids in a peptide chain
Electronics	Communication Protocol (I2C, SPI, UART, CAN bus etc.)	Electrical parameters like Voltage,current,resistance etc.
Mechanical	Failure modes (Wear, yielding, corrosion etc.)	Drilling data like ROP, Klogh, bit depth etc.



- **Raw vs Processed Data**

  Data can also be classified based on its stage of readiness for analysis, namely raw and processed. Raw data is information collected directly from an instrument, sensor or experiment without any modification. It may contain noise, missing values or inconsistencies introduced during measurement or recording. Processed data is raw data that has been cleaned, corrected and structured so it can be used for analysis. This may involve removing erroneous values, filling missing entries, converting units to a consistent scale or aggregating readings taken at different times.

  For example, a raw voltage log from a circuit may contain sudden spikes caused by sensor glitches whereas a processed version of the same log would have those spikes filtered out or corrected. Similarly, a biochemistry dataset may have blank entries where a reading failed and these would be filled or removed during processing. The distinction matters because conclusions drawn directly from raw data risk being skewed by errors that have not yet been identified or corrected. Processing is therefore a required step between data collection and meaningful analysis rather than an optional one.

Measurement Units and their Importances

Every numerical values in a dataset is only meaningful when it is paired with its correct unit of measurement. A numerical value on its own contains no physical meaning, The unit tells us what the value actually represents and allows the value to be correctly interpreted within its domain.

Comparing values from different units directly can lead to incorrect conclusions since the values may represent entirely different physical quantities despite having similar magnitudes. For example a pH reading of 7 and a voltage reading of 7 volts are numerically identical but represent completely unrelated measurements and cannot be compared or combined in any meaningful way. Similarly 5 volts and 5 milliamps both use the number 5 but refer to voltage and current respectively, which are different electrical quantities. Before analyzing any dataset it is necessary to confirm the unit of each variable and ensure that all values within a variable are expressed in a consistent unit before any comparison or calculation is performed.

Summary Statistics

Basic summary statistics provide a quick understanding of a dataset before deeper analysis is carried out. Mean is the average of all values in a dataset, calculated by summing the values and dividing by the total count. Median is the middle value of a dataset when all values are arranged in order and is less affected by extreme values than the mean.

Minimum and maximum represent the smallest and largest values present in the dataset while range is the difference between the two and gives a sense of how spread out the data is. Together these statistics offer a first impression of a dataset without requiring any advanced analysis.

The mean can be misleading when a dataset contains outliers or is heavily skewed. For instance if most drilling depth readings fall within a normal range but one erroneous sensor reading is unusually high, the mean will be pulled upward even though it does not reflect the majority of the data. In such cases the median gives a more accurate sense of the typical value in the dataset.

Pre-Analysis Considerations

Before analyzing any dataset certain checks must be carried out to avoid drawing incorrect conclusions. This includes confirming the unit of each variable, checking for missing or null values and verifying that data types match what is expected for each column. Identifying outliers and understanding the experimental condition under which the data was collected are equally important at this stage.

When receiving domain data from another team it is equally important to ask clarifying questions before beginning analysis. Relevant questions include what unit each variable is measured in, whether the data is raw or already processed and under what condition or setup the data was recorded. Asking what range of values is considered normal for that variable is also useful, since without this context even a technically correct analysis can lead to conclusions that are not valid for the domain.

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